In [1]:
```
# importing pandas and seaborn module
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
#plt.style.use('fivethirtyeight')
import warnings
warnings.filterwarnings('ignore')   #this will ignore the warnings.it wont display
```

In [2]:
```
iris=pd.read_csv(r"C:\Users\Aisharani Patro\OneDrive\Desktop\NIT\MARCH\25th - Seabo
iris
```

Out[2]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [3]:
```
iris.head()
```

Out[3]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [4]:
```
iris.drop('Id',axis=1,inplace=True)
```

In [5]:
```
iris.head()
```

Out[5]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [6]:
```
iris.info()
```

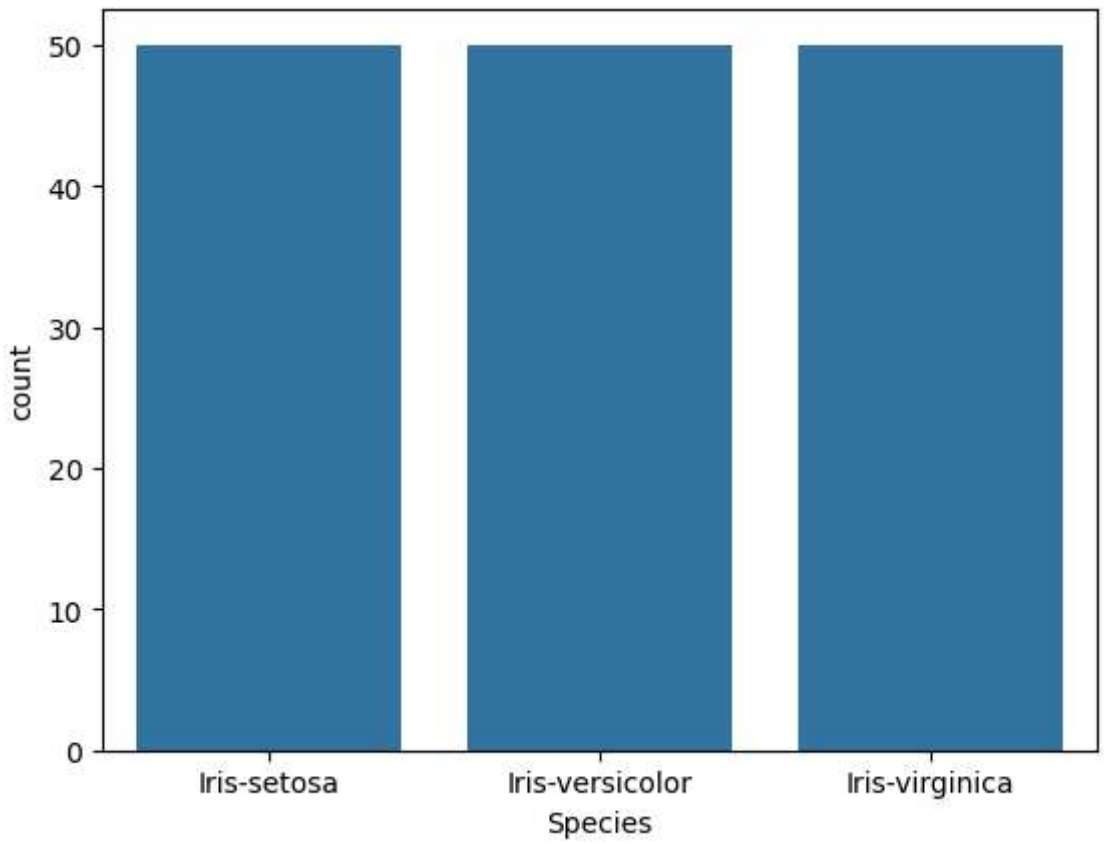
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   SepalLengthCm  150 non-null    float64
 1   SepalWidthCm   150 non-null    float64
 2   PetalLengthCm  150 non-null    float64
 3   PetalWidthCm   150 non-null    float64
 4   Species        150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [7]:
```
iris['Species'].value_counts()
```

Out[7]:
```
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

Bar Plot

In [8]:
```
sns.countplot(x='Species',data=iris)
plt.show()
```



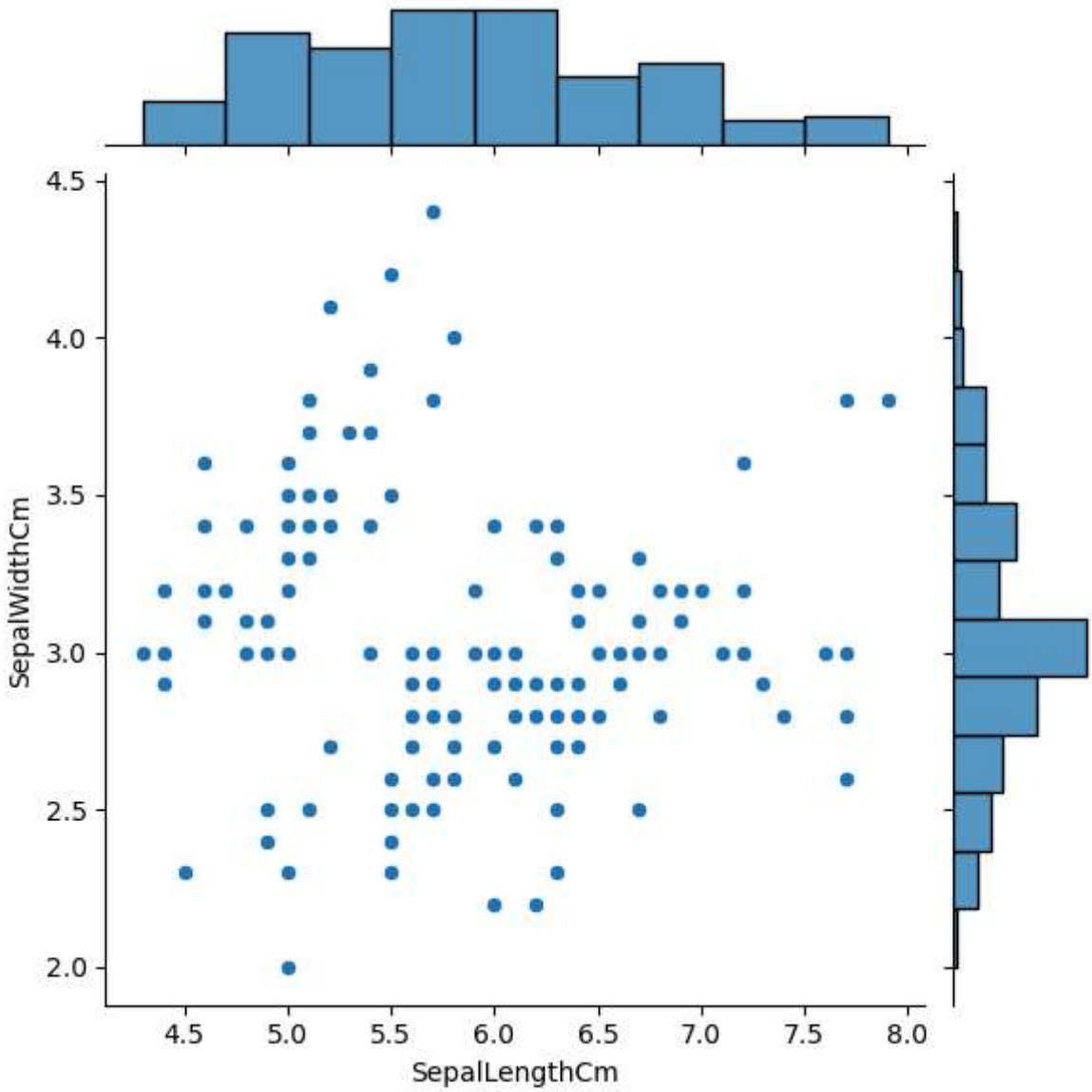
Joint Plot

In [9]:
```
iris.head()
```

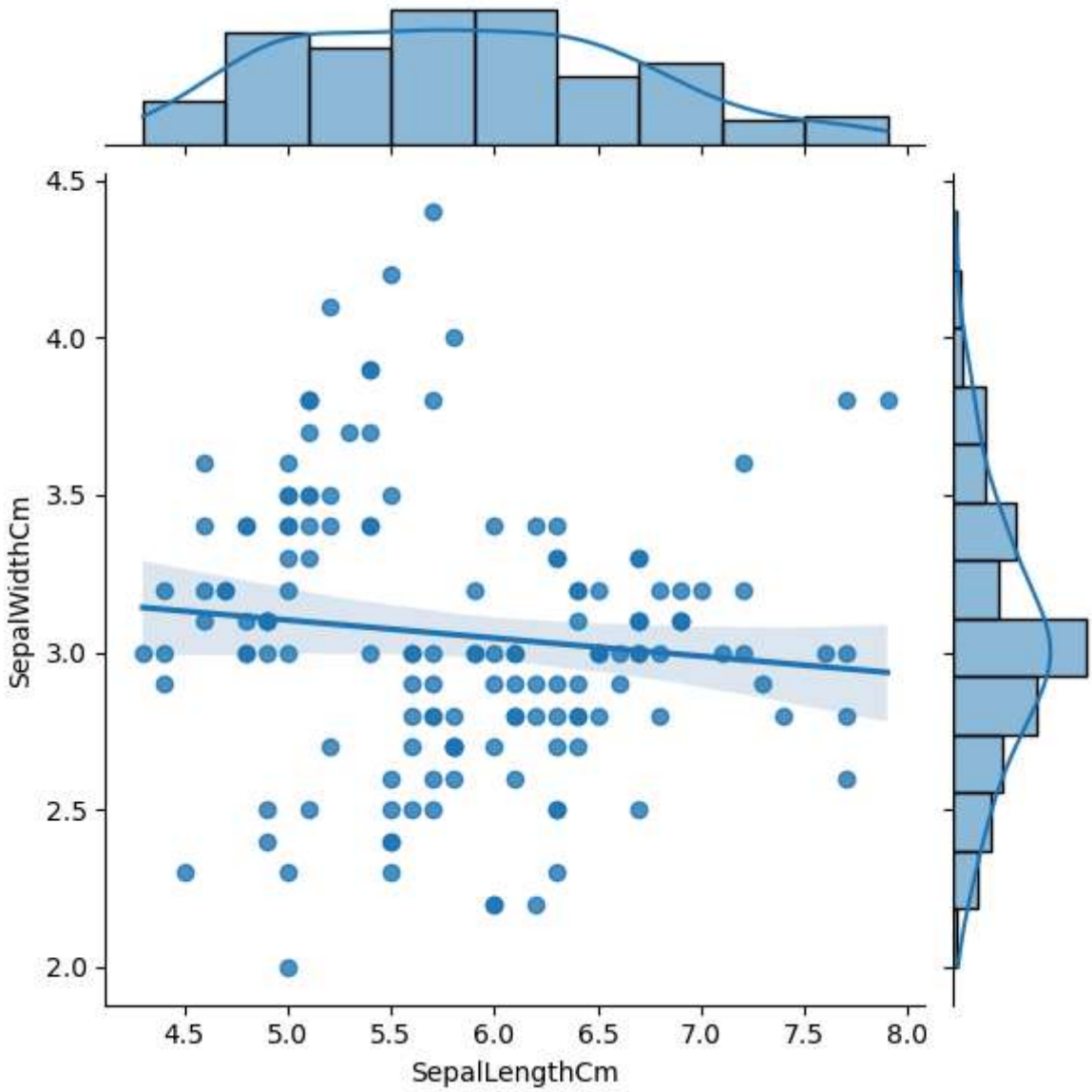
Out[9]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

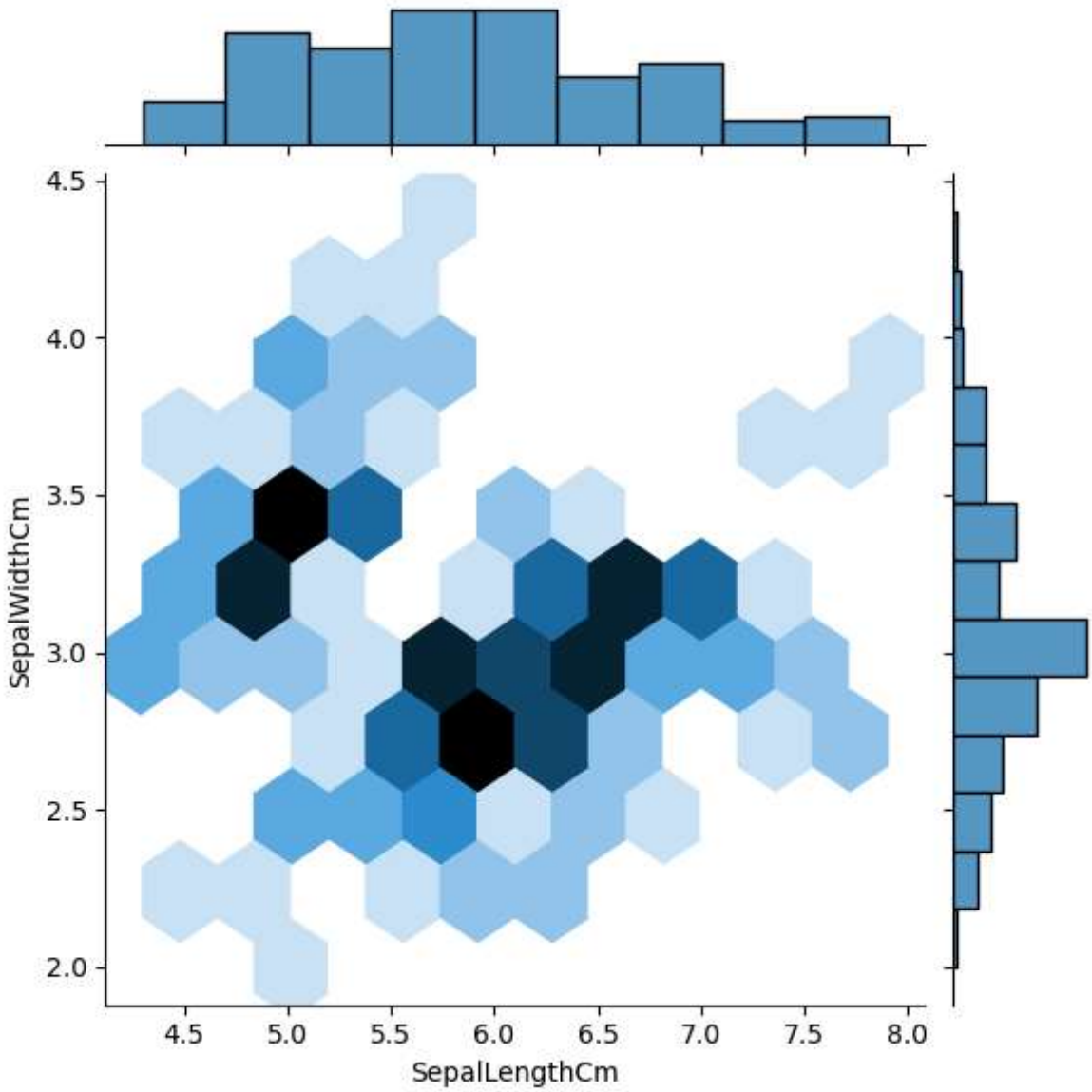
In [10]:
```
fig=sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',data=iris)
```



In [11]:
```
fig=sns.jointplot(x="SepalLengthCm",y="SepalWidthCm",data=iris,kind="reg")
```



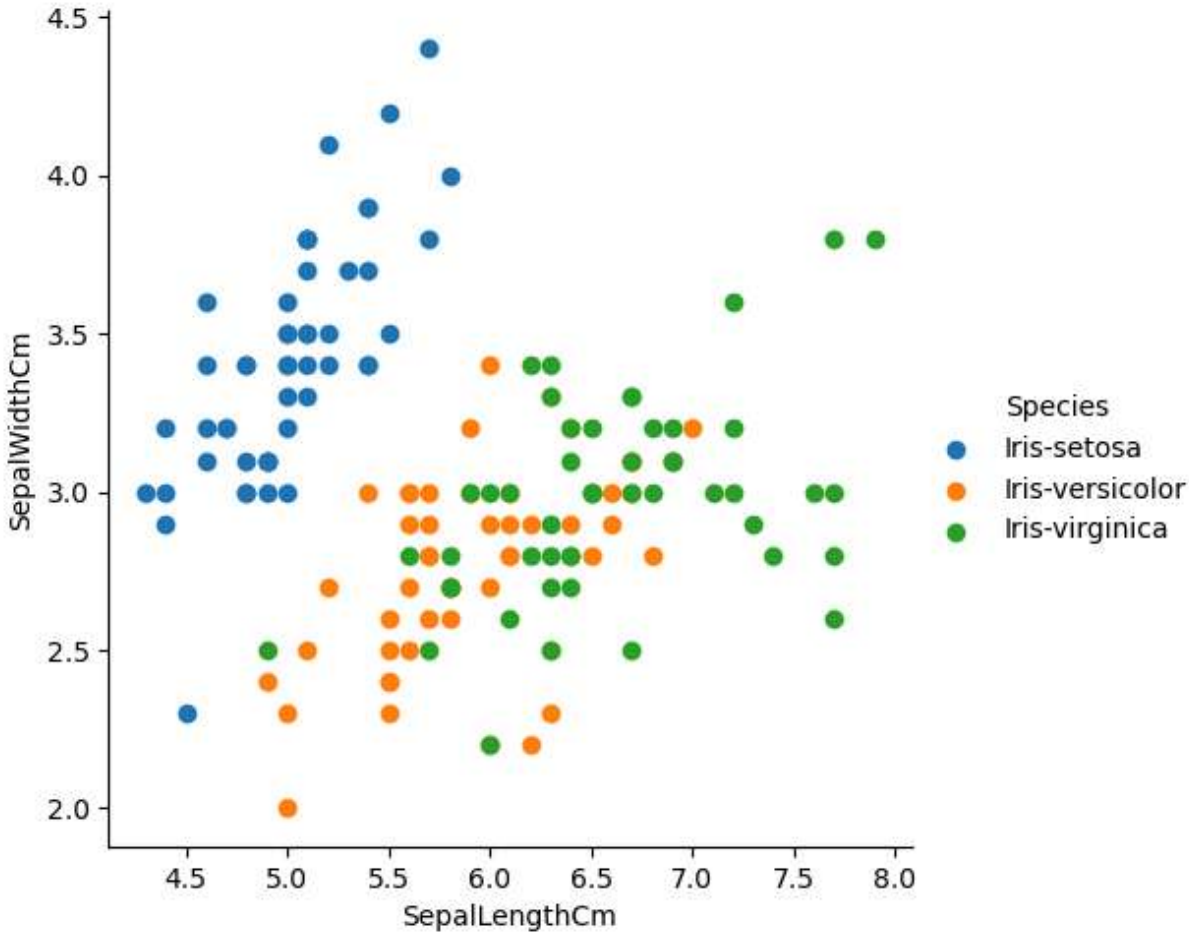
In [12]:
```
fig=sns.jointplot(x='SepalLengthCm',y='SepalWidthCm',kind='hex',data=iris)
```



FacetGrid Plot

In [13]:
```
import matplotlib.pyplot as plt
%matplotlib inline

sns.FacetGrid(iris,hue='Species',height=5)\
.map(plt.scatter,'SepalLengthCm','SepalWidthCm')\
.add_legend()
```

Out[13]:
```
<seaborn.axisgrid.FacetGrid at 0x1f5738eb770>
```



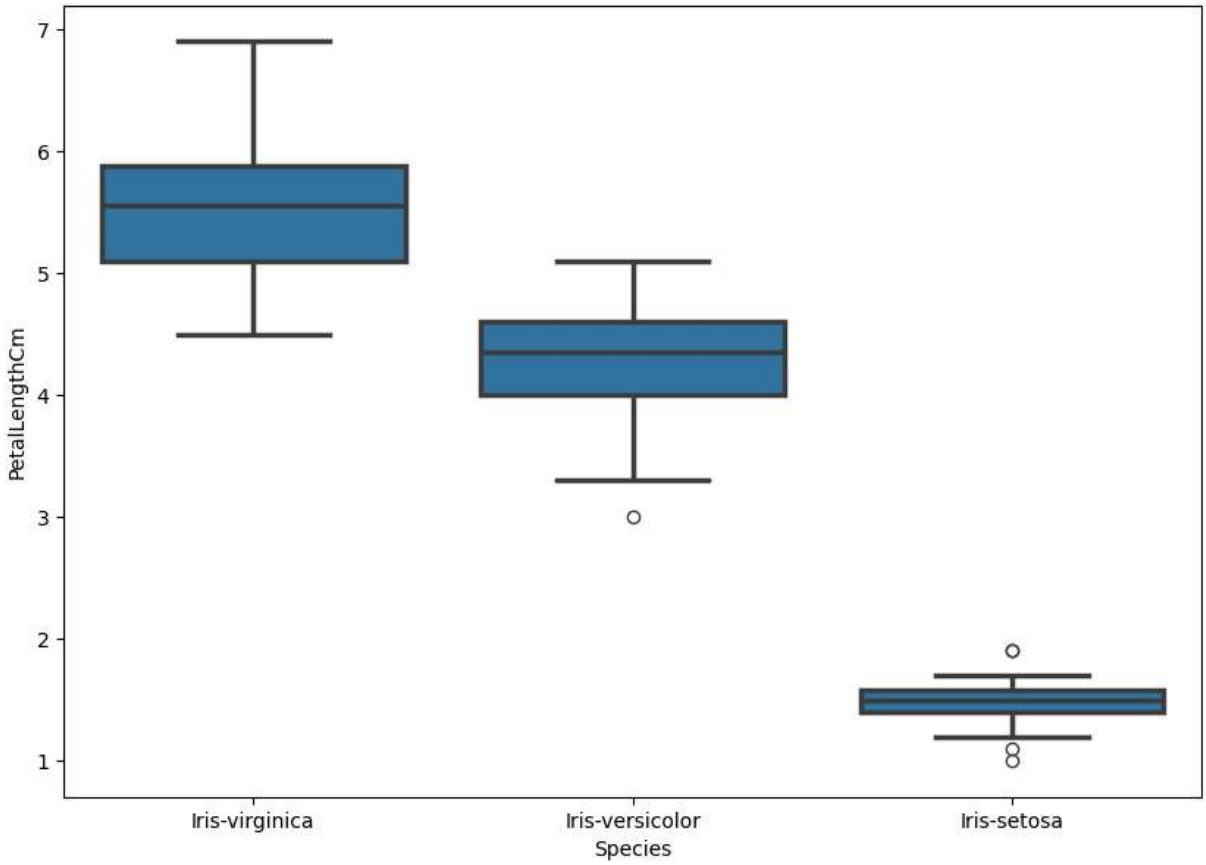
Boxplot or Whisker plot

In [14]:
```
iris.head()
```

Out[14]:

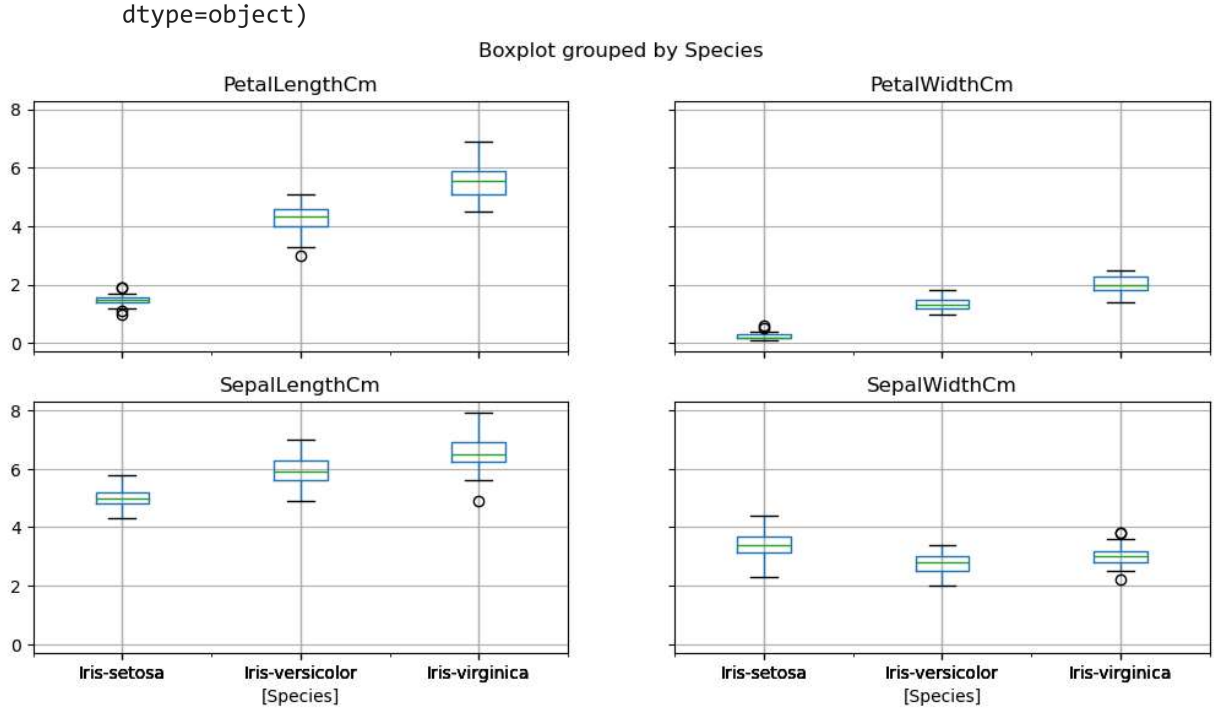
	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [15]:
```
fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.boxplot(x='Species',y='PetalLengthCm',data=iris,order=['Iris-virginica','Ir
```



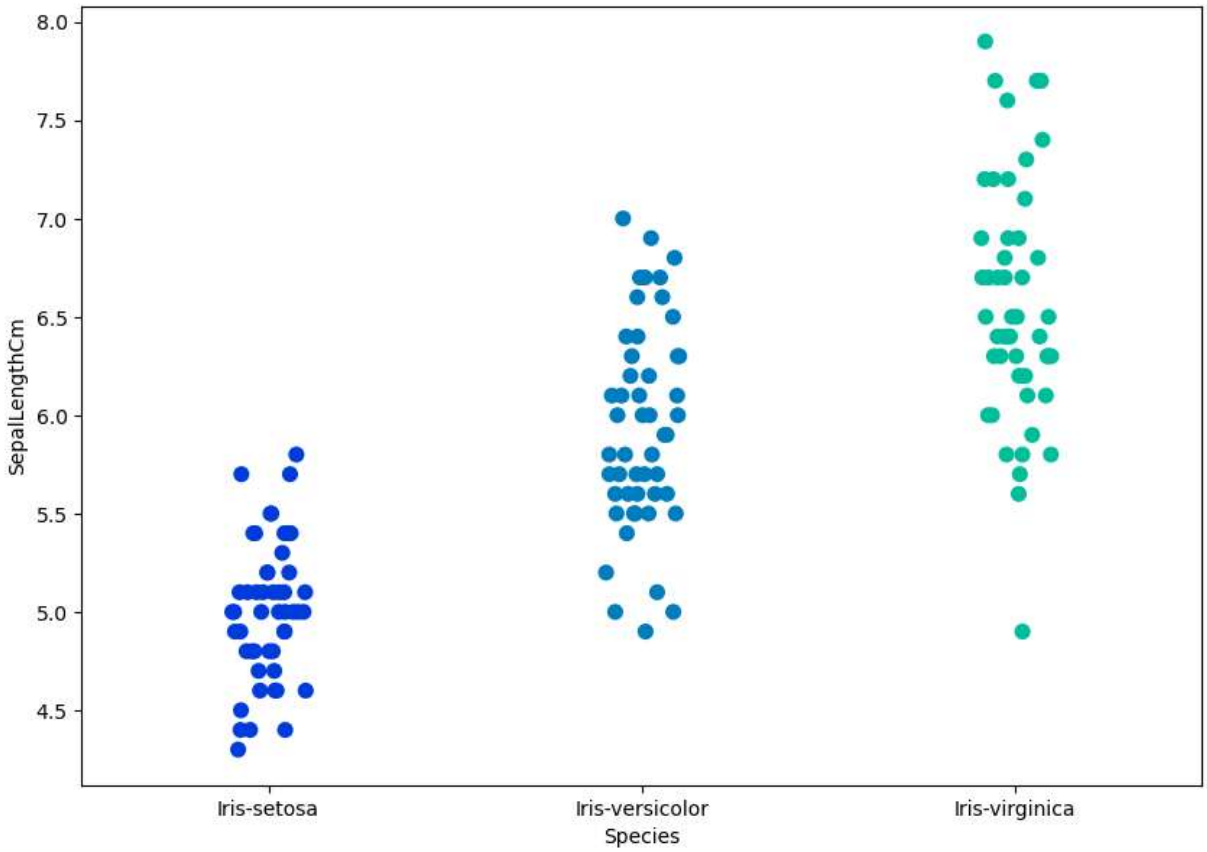
In [16]:
```
#iris.drop("Id",axis=1).boxplot(by="Species",figsize=(12,6))
iris.boxplot(by="Species",figsize=(12,6))
```

Out[16]:
```
array([[<Axes: title={'center': 'PetalLengthCm'}, xlabel='[Species]'>,
        <Axes: title={'center': 'PetalWidthCm'}, xlabel='[Species]'>],
       [<Axes: title={'center': 'SepalLengthCm'}, xlabel='[Species]'>,
        <Axes: title={'center': 'SepalWidthCm'}, xlabel='[Species]'>]],
      dtype=object)
```



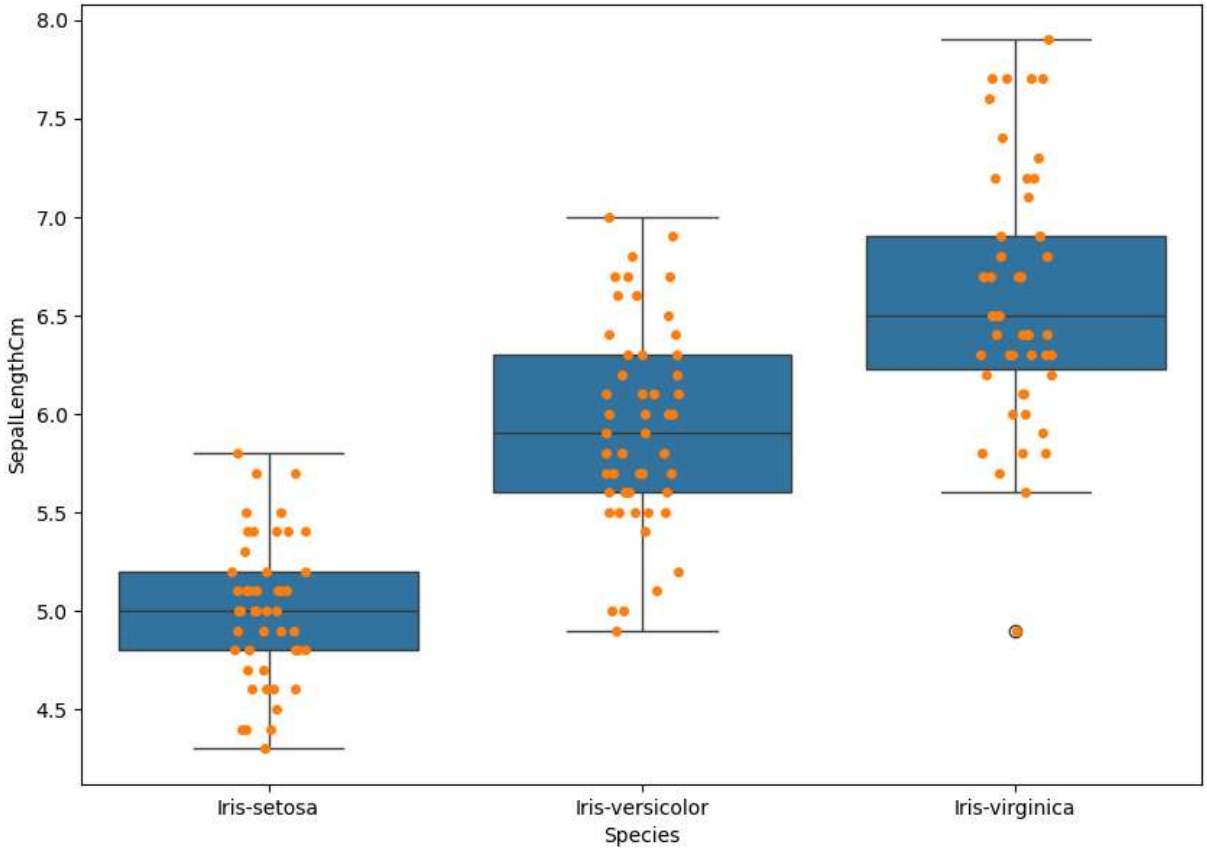
Strip plot

In [17]:
```
fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.stripplot(x='Species',y='SepalLengthCm',data=iris,jitter=True,edgecolor='gr
```

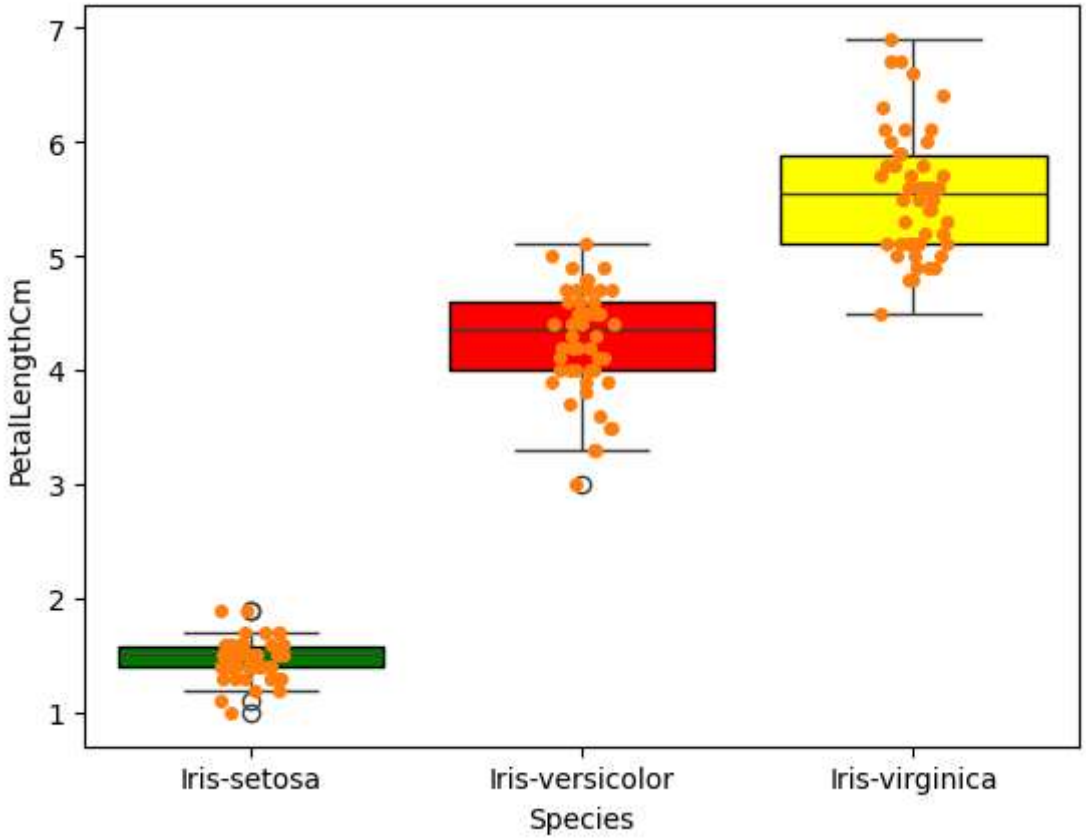


Combining Box and Strip Plots

In [18]:
```
fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.boxplot(x='Species',y='SepalLengthCm',data=iris)
fig=sns.stripplot(x='Species',y='SepalLengthCm',data=iris,jitter=True,edgecolor='gr
```

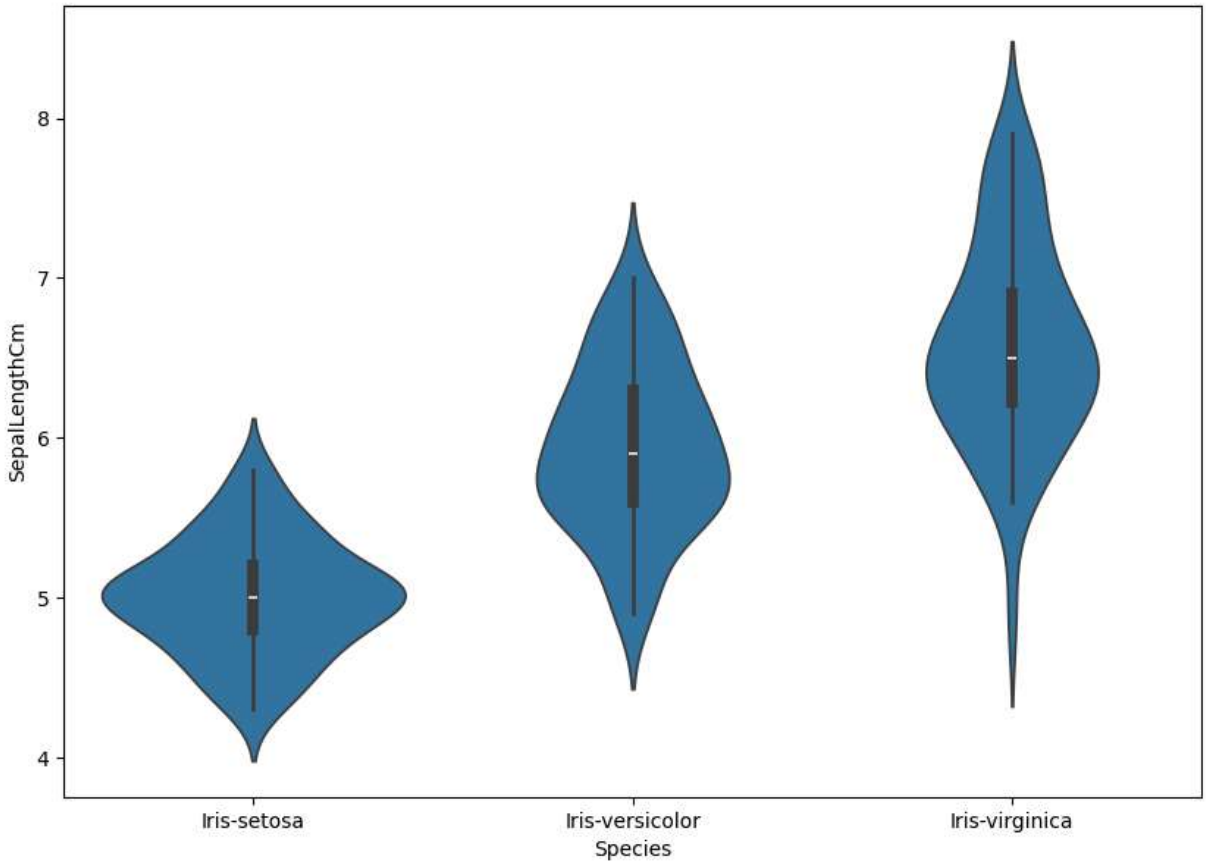


In [19]:
```
ax=sns.boxplot(x="Species",y="PetalLengthCm",data=iris)
ax=sns.stripplot(x="Species",y="PetalLengthCm",data=iris,jitter=True,edgecolor='gre

boxtwo=ax.patches[2]
boxtwo.set_facecolor('yellow')
boxtwo.set_edgecolor('black')
boxthree=ax.patches[1]
boxthree.set_facecolor('red')
boxthree.set_edgecolor('black')
boxthree=ax.patches[0]
boxthree.set_facecolor('green')
boxthree.set_edgecolor('black')

plt.show()
```



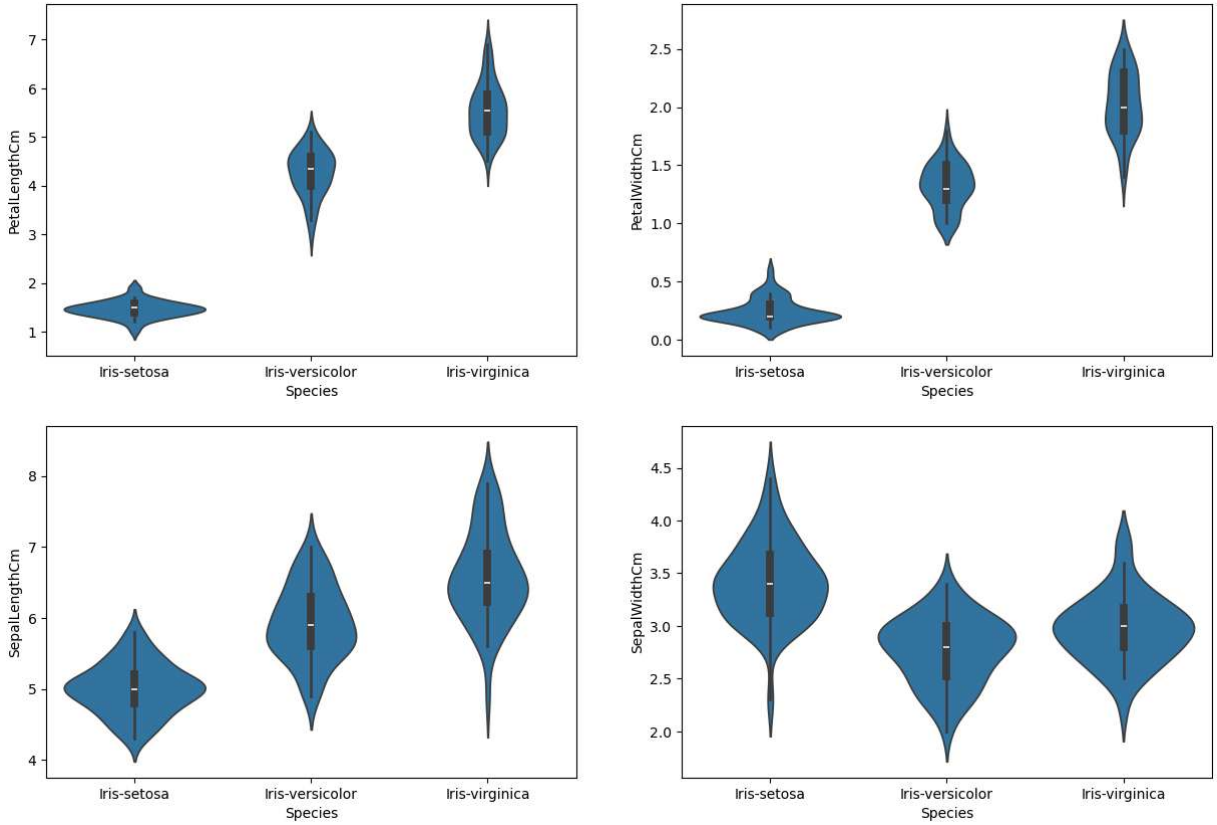
Violin Plot

In [20]:
```
fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.violinplot(x="Species",y="SepalLengthCm",data=iris)
```



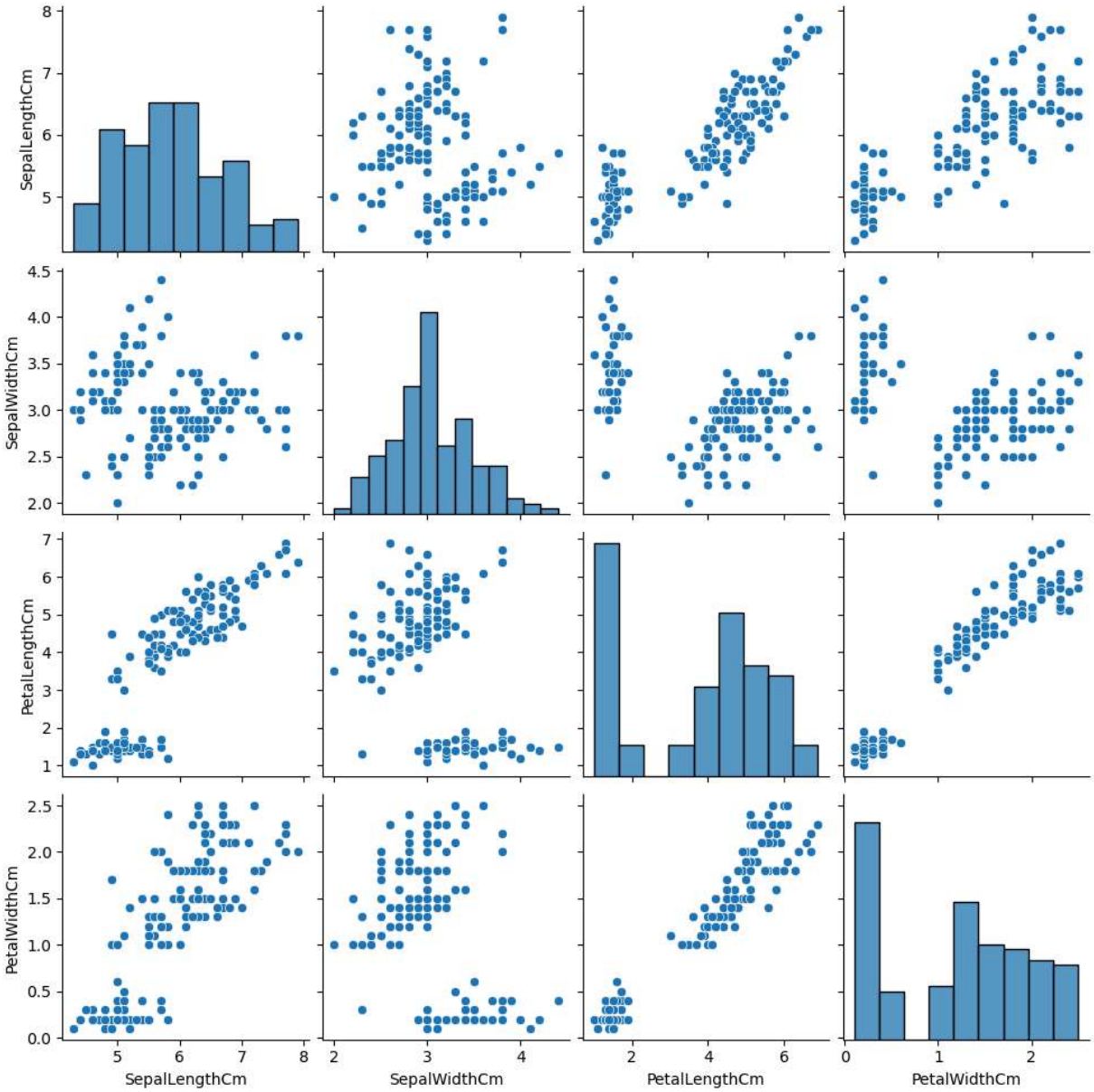
In [21]:
```
plt.figure(figsize=(15,10))
plt.subplot(2,2,1)
sns.violinplot(x="Species",y="PetalLengthCm",data=iris)
plt.subplot(2,2,2)
sns.violinplot(x="Species",y="PetalWidthCm",data=iris)
plt.subplot(2,2,3)
sns.violinplot(x="Species",y="SepalLengthCm",data=iris)
plt.subplot(2,2,4)
sns.violinplot(x="Species",y="SepalWidthCm",data=iris)
```

Out[21]: <Axes: xlabel='Species', ylabel='SepalWidthCm'>

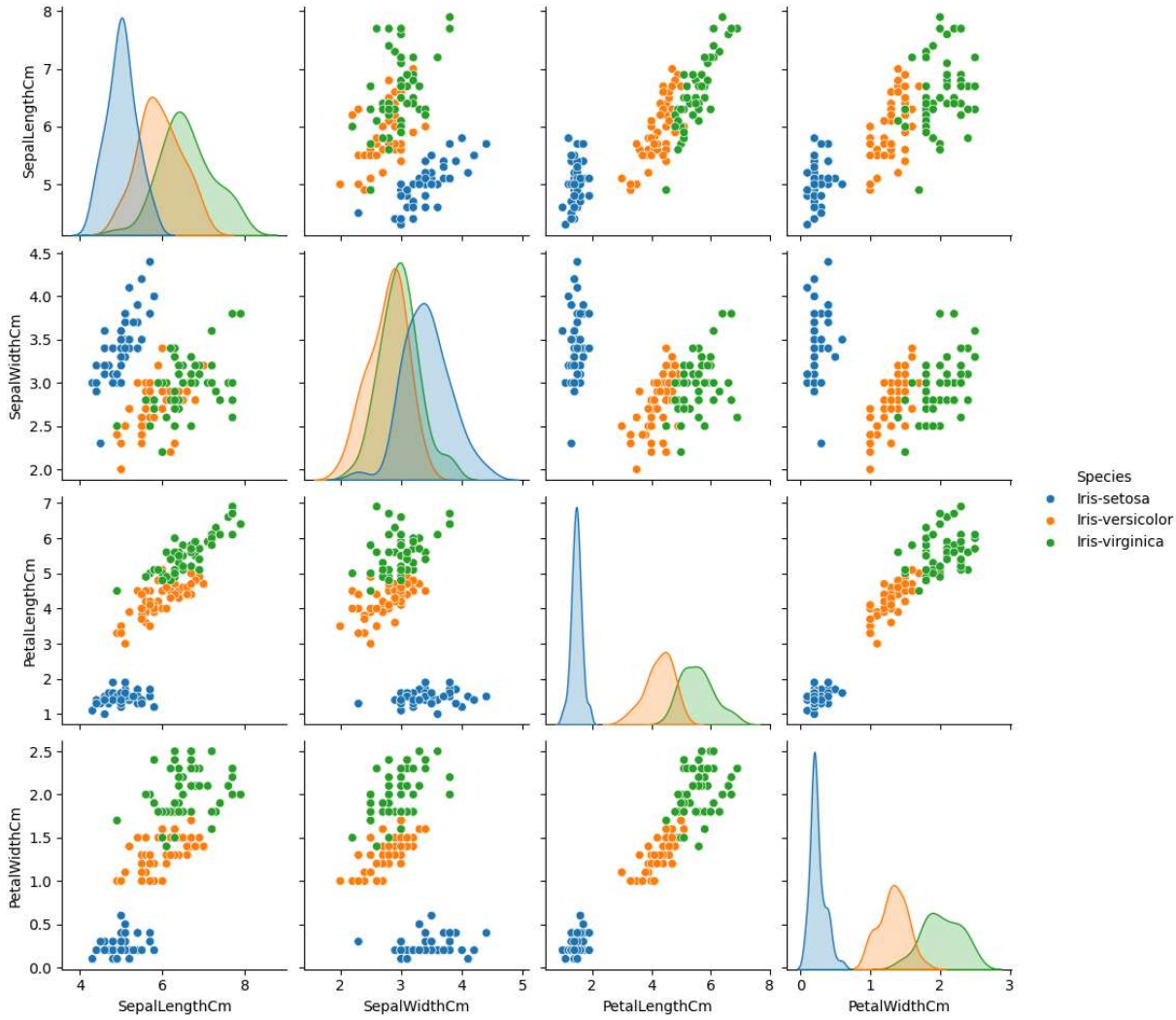


Pair plot

In [22]:
```
fig=sns.pairplot(data=iris,kind='scatter')
```

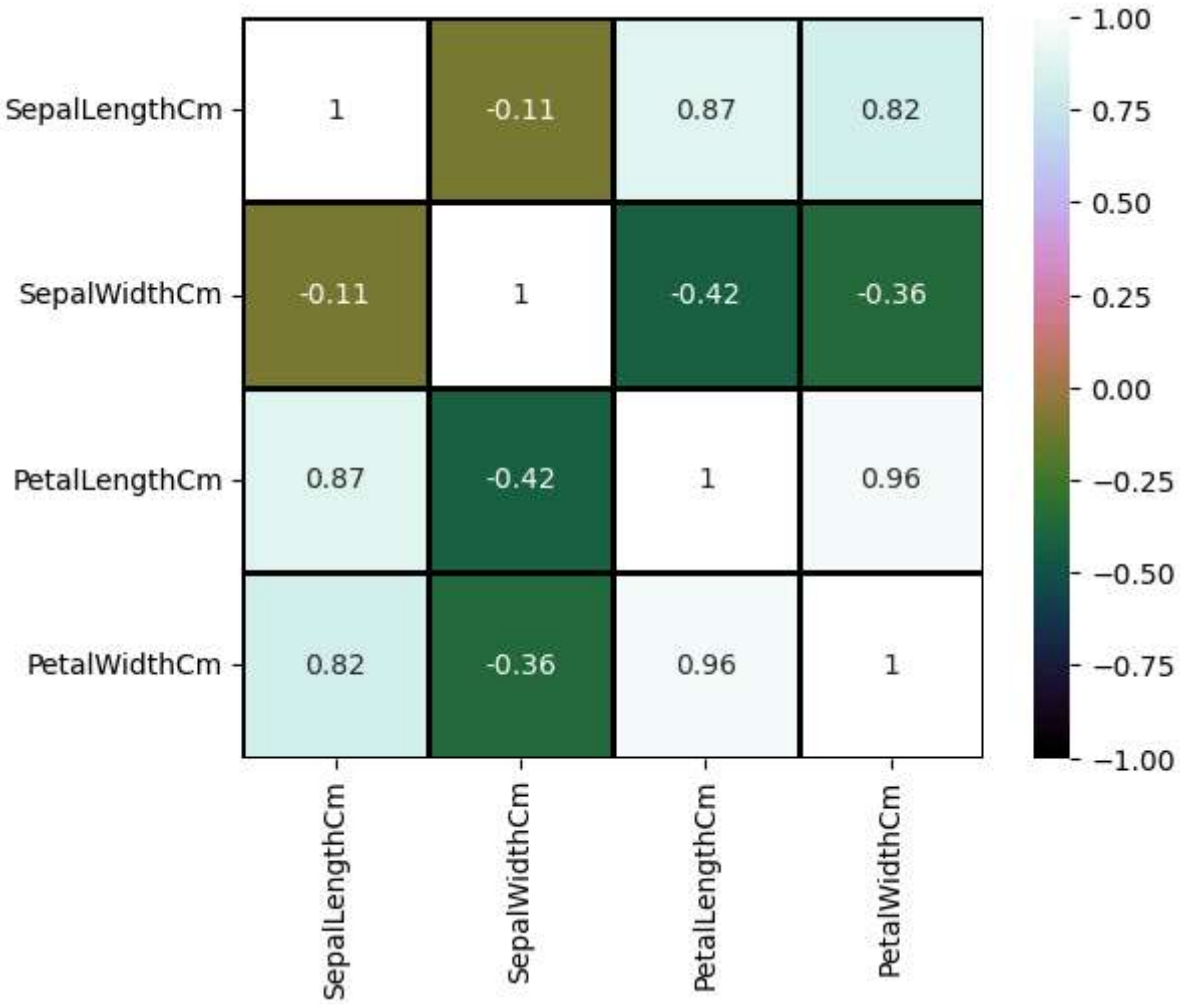


In [23]:
```
fig=sns.pairplot(iris,hue='Species')
```



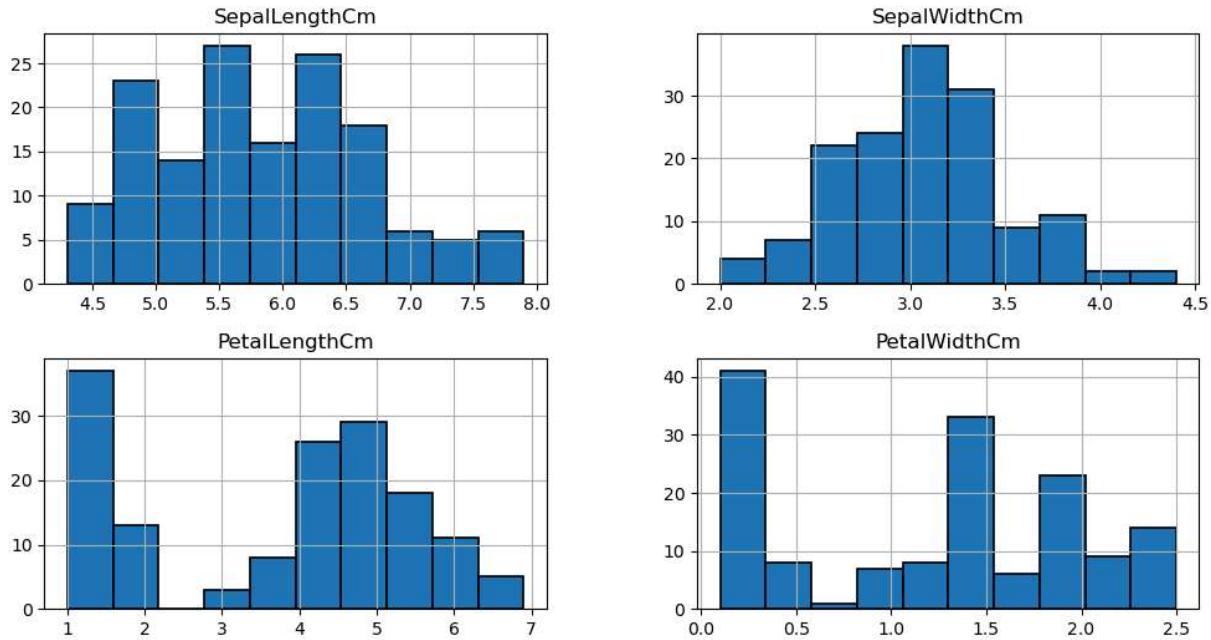
Heat Map

In [24]:
```
fig=sns.heatmap(iris.corr(numeric_only=True),annot=True,cmap='cubehelix',linewidths
```

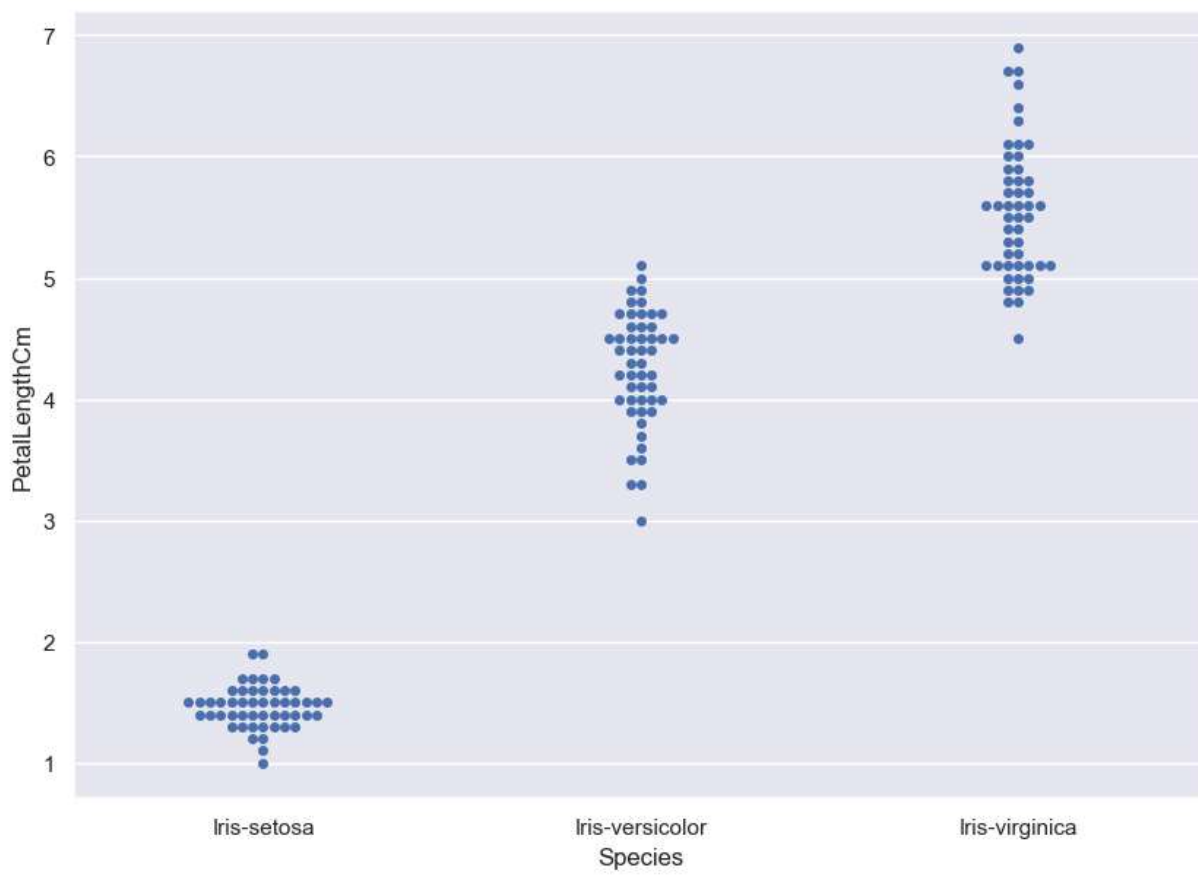


Distribution plot

In [25]:
```
iris.hist(edgecolor='black',linewidth=1.2)
fig=plt.gcf()
fig.set_size_inches(12,6)
```

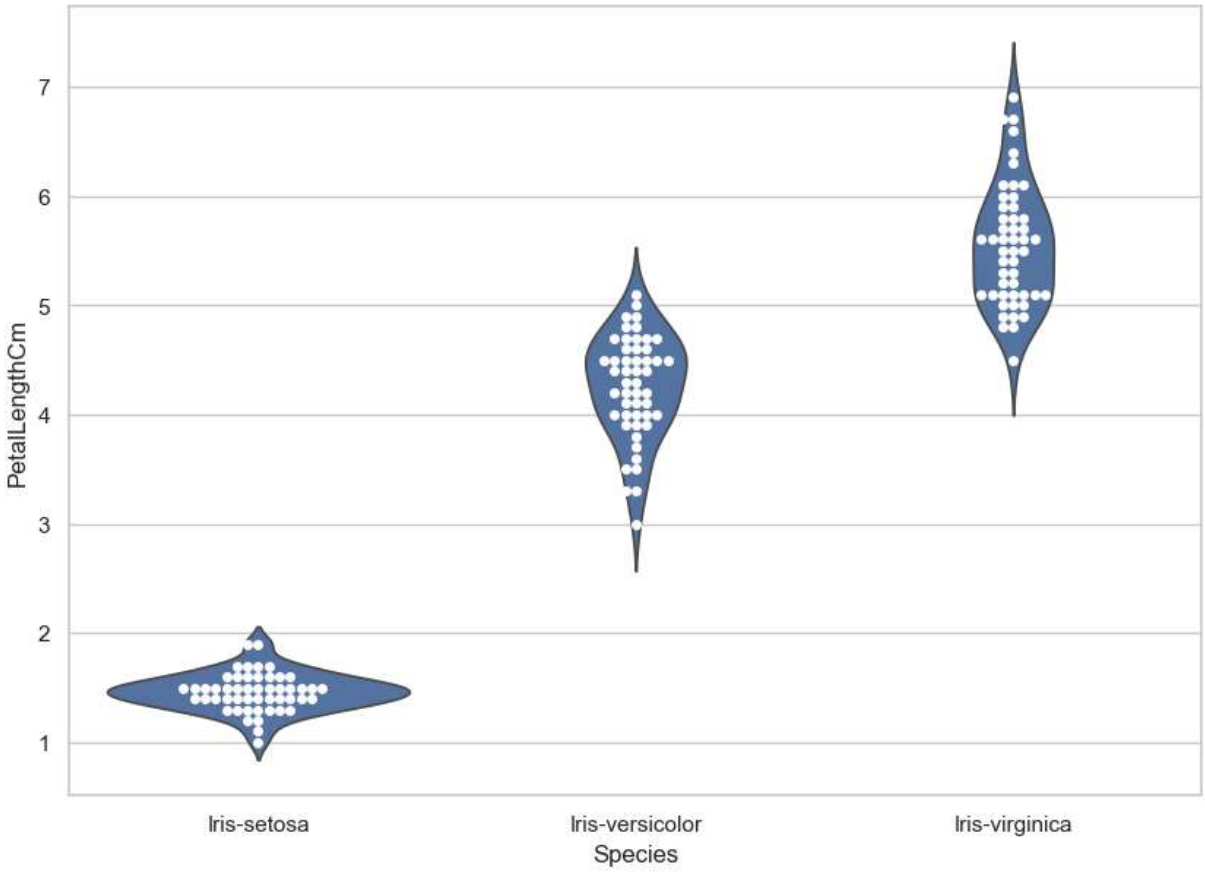


In [26]:
```
sns.set(style="darkgrid")
fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.swarmplot(x="Species",y="PetalLengthCm",data=iris)
```



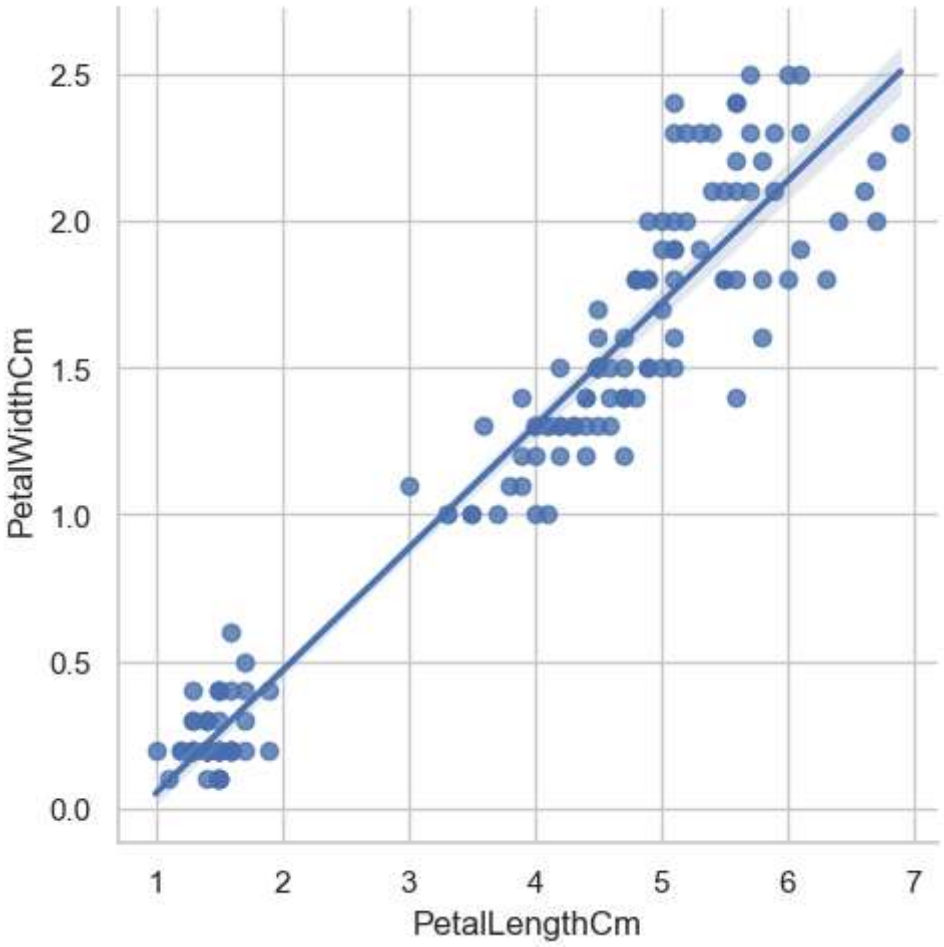
In [27]:
```
sns.set(style="whitegrid")
fig=plt.gcf()
fig.set_size_inches(10,7)
```

```
ax=sns.violinplot(x="Species",y="PetalLengthCm",data=iris,inner=None)
ax=sns.swarmplot(x="Species",y="PetalLengthCm",data=iris,color="white",edgecolor="b
```



LM plot

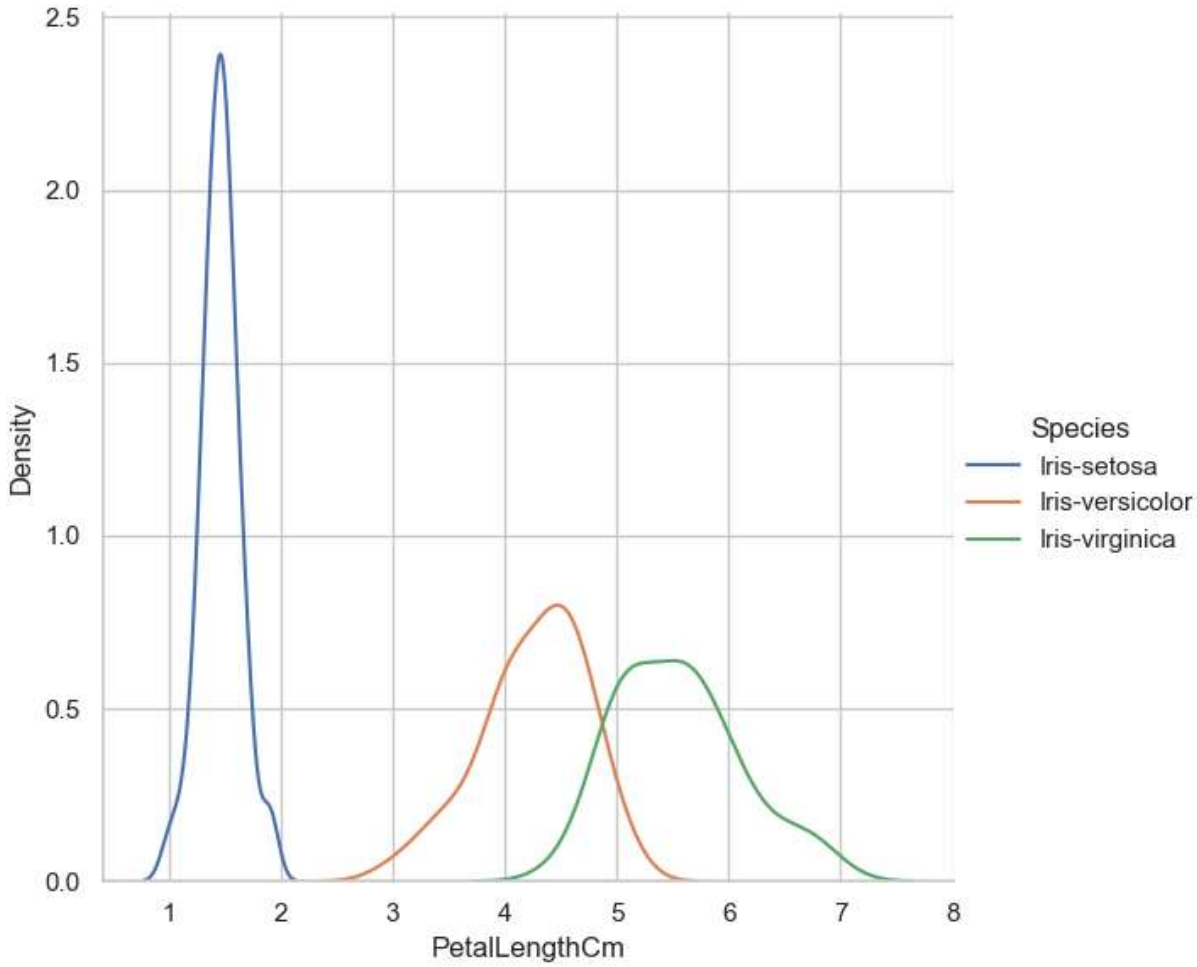
In [28]:
```
fig=sns.lmplot(x="PetalLengthCm",y="PetalWidthCm",data=iris)
```



Facet Grid

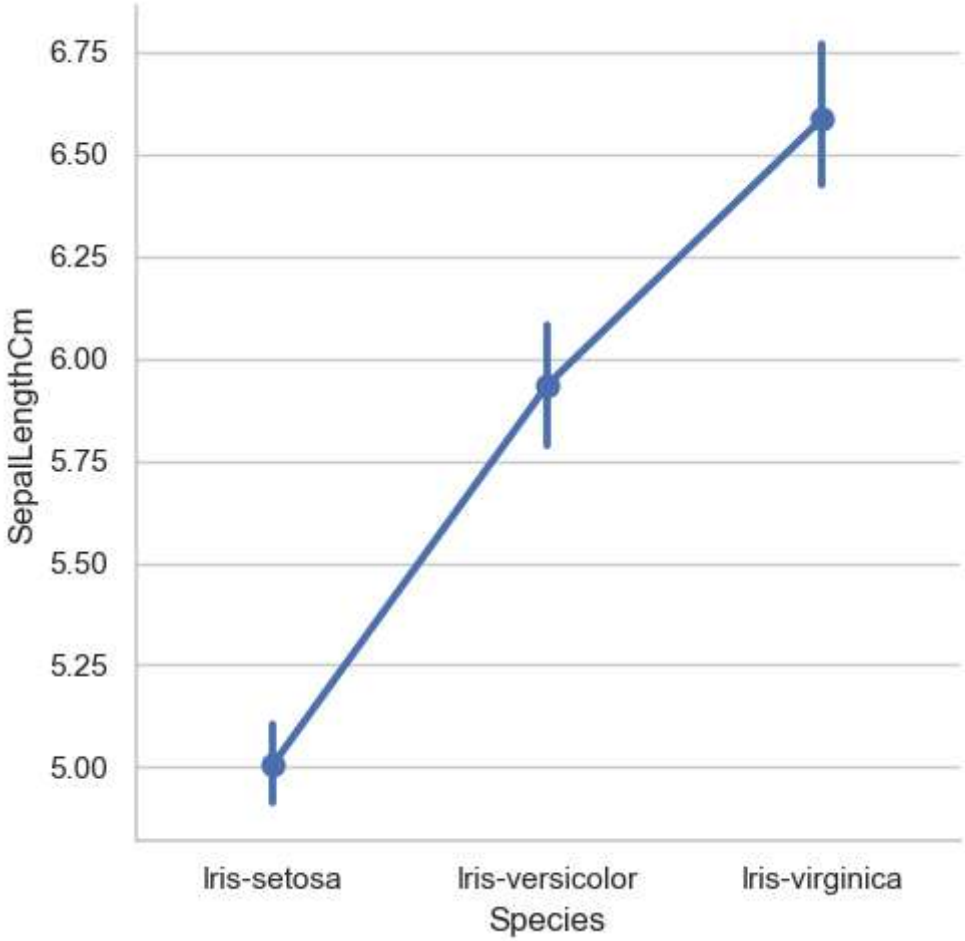
In [29]:
```
sns.FacetGrid(iris,hue="Species",height=6) \
    .map(sns.kdeplot,"PetalLengthCm") \
    .add_legend()
plt.ioff()
```

Out[29]:
```
<contextlib.ExitStack at 0x1f57dedc5a0>
```



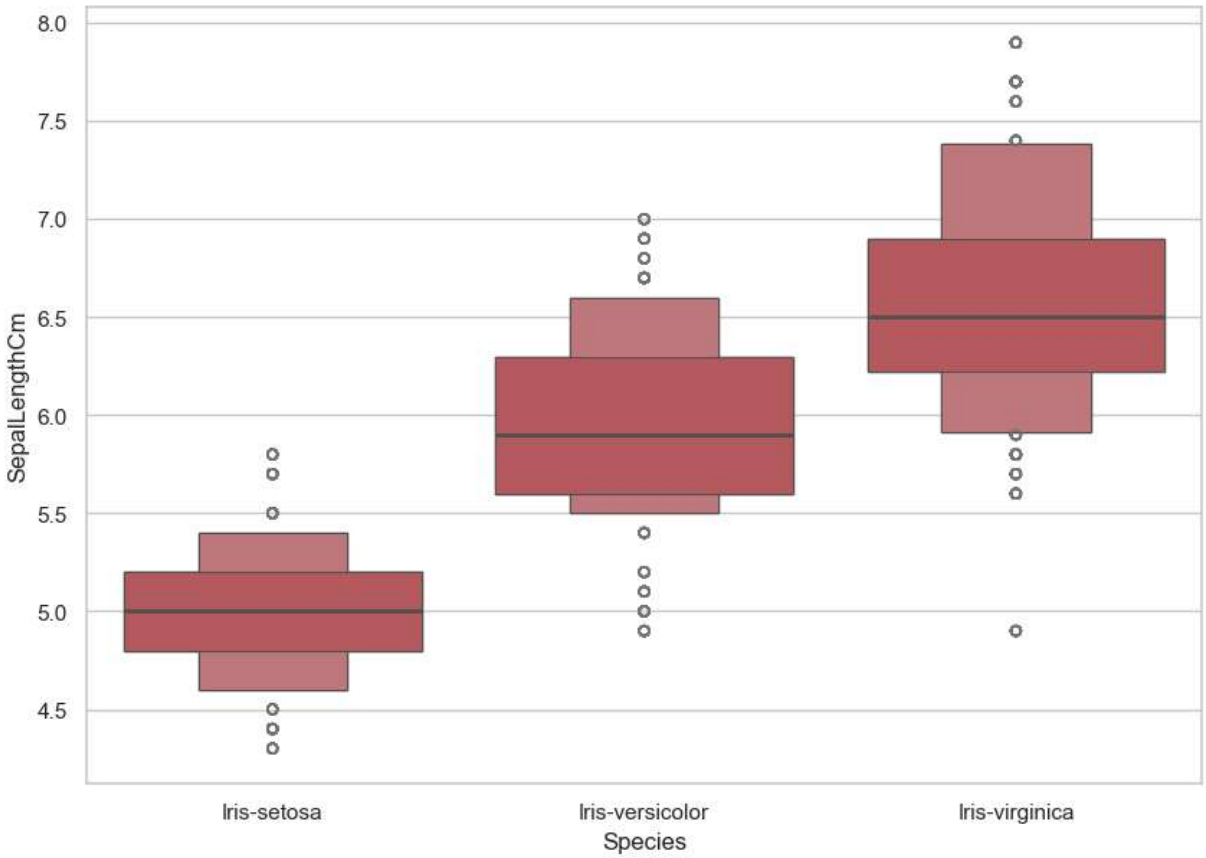
Catplot

In [30]:
```
#f,ax=plt.subplots(1,2,figsize=(18,8))
sns.catplot(x='Species',y='SepalLengthCm',data=iris,kind='point')
plt.ioff()
plt.show()
#sns.catplot('Species','SepalLengthCm',data=iris,ax=ax[0][0])
#sns.catplot('Species','SepalWidthCm',data=iris,ax=ax[0][1])
#sns.catplot('Species','PetalLengthCm',data=iris,ax=ax[1][0])
#sns.catplot('Species','PetalWidthCm',data=iris,ax=ax[1][1])
```



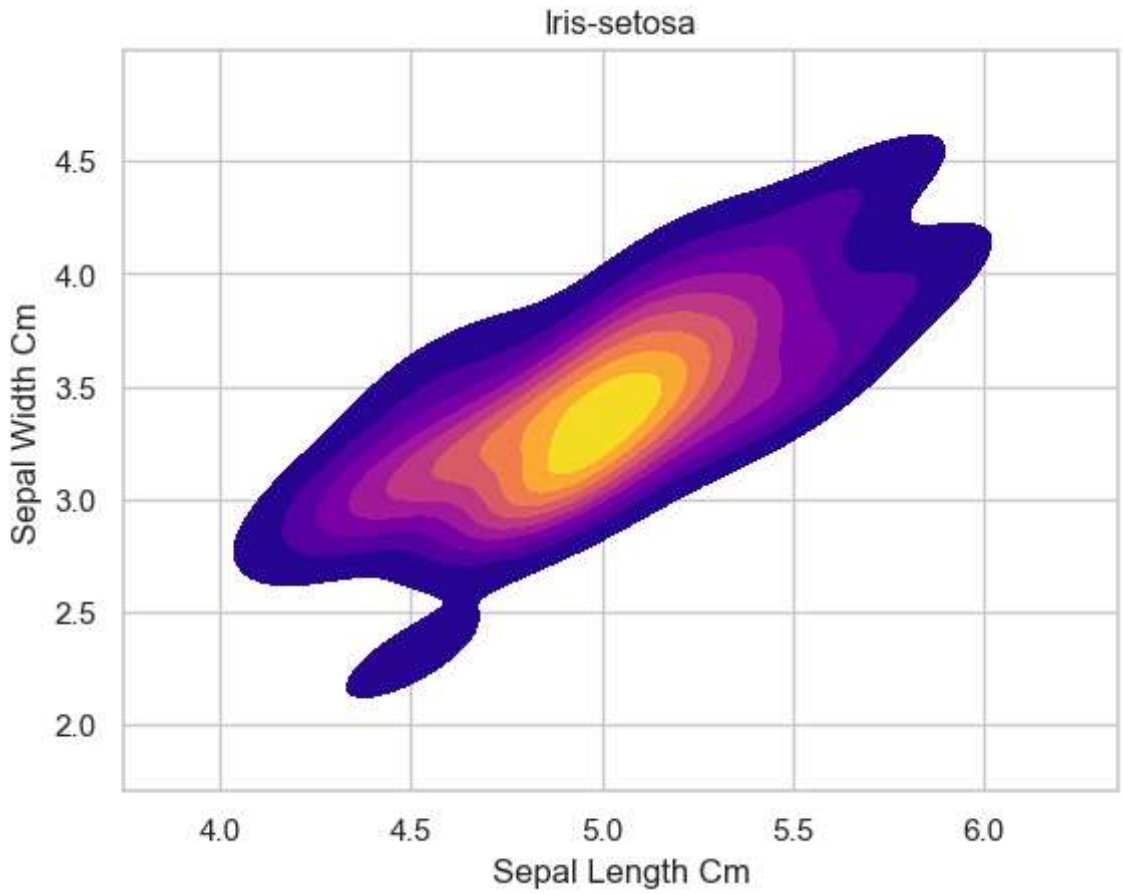
Boxen Plot

In [44]:
```
fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.boxenplot(x='Species',y='SepalLengthCm',data=iris)
plt.show()
```



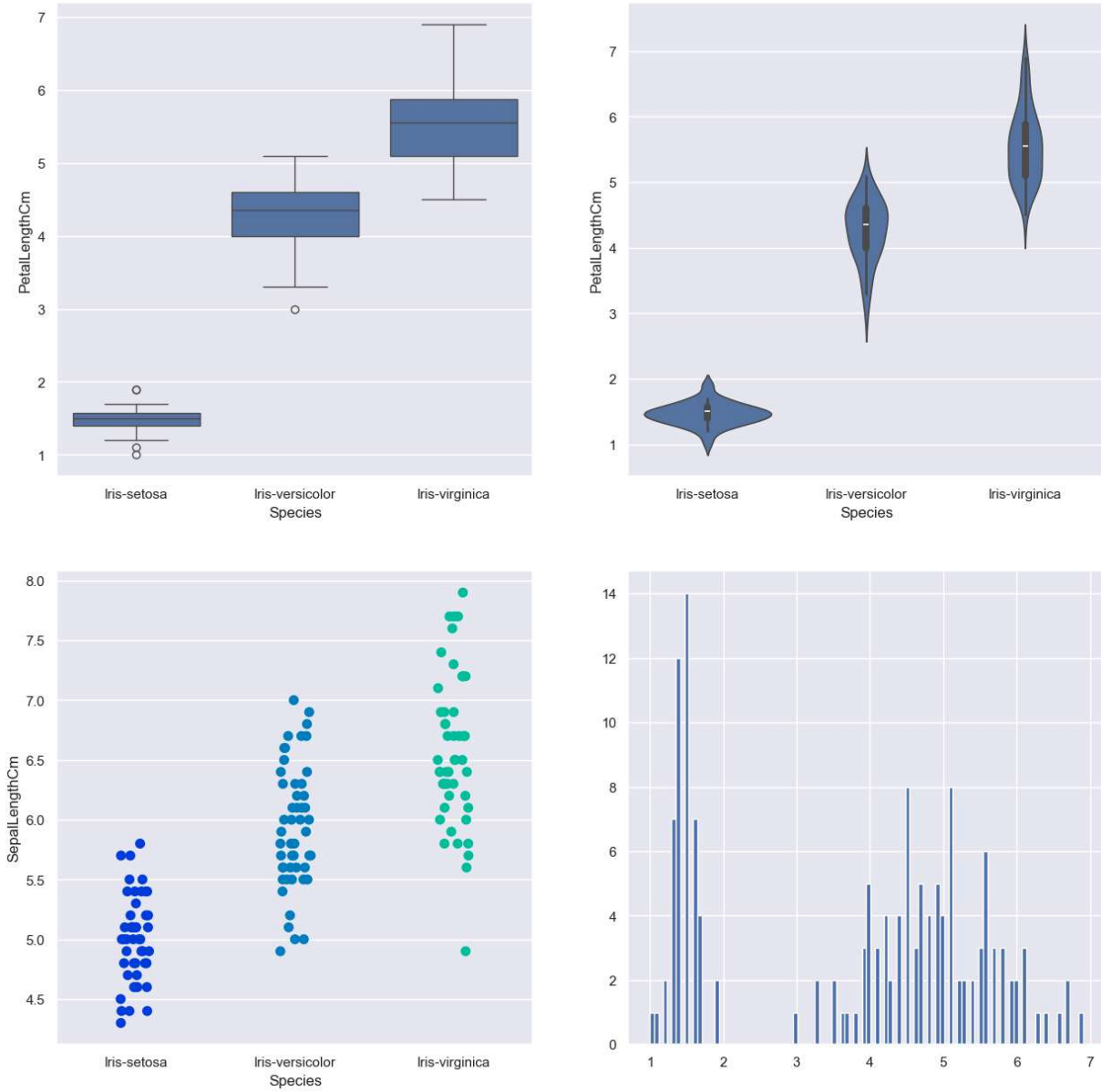
KDE Plot

In [50]:
```
sub=iris[iris['Species']=='Iris-setosa']
sns.kdeplot(x=sub['SepalLengthCm'],y=sub['SepalWidthCm'],cmap="plasma",fill=True)
plt.title('Iris-setosa')
plt.xlabel('Sepal Length Cm')
plt.ylabel('Sepal Width Cm')
plt.show()
```



Dashboard

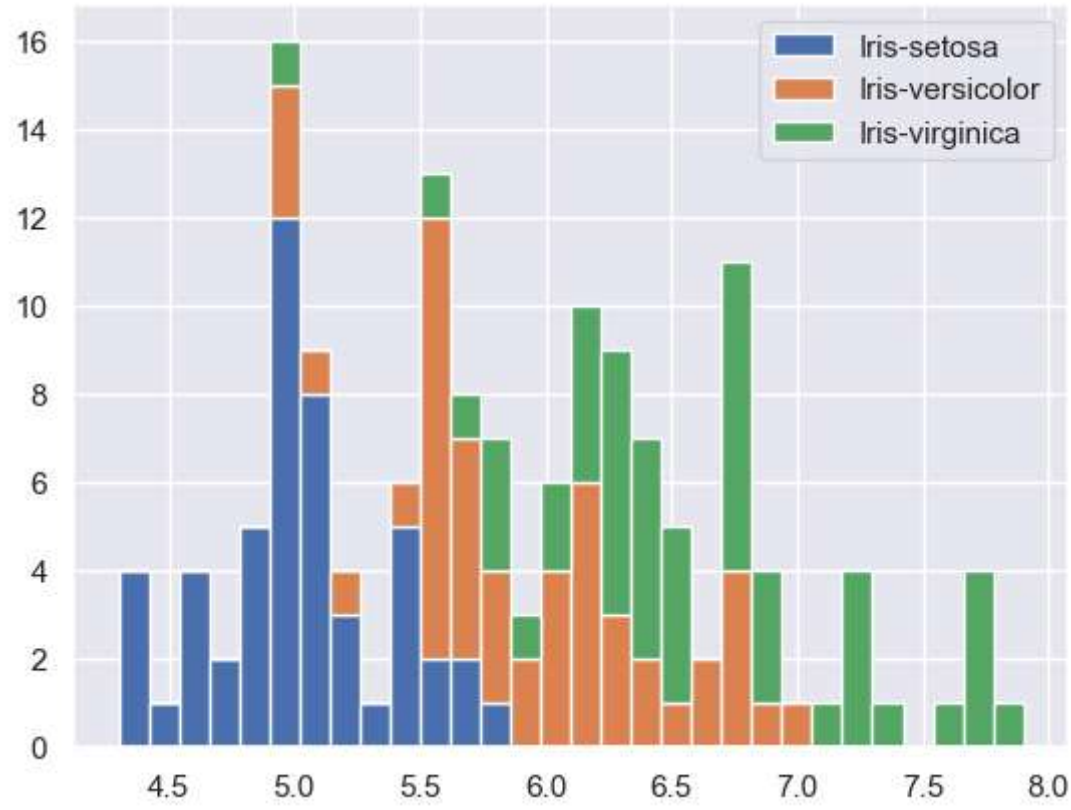
In [51]:
```
sns.set_style('darkgrid')
f,axes=plt.subplots(2,2,figsize=(15,15))

k1=sns.boxplot(x="Species", y="PetalLengthCm", data=iris,ax=axes[0,0])
k2=sns.violinplot(x='Species',y='PetalLengthCm',data=iris,ax=axes[0,1])
k3=sns.stripplot(x='Species',y='SepalLengthCm',data=iris,jitter=True,edgecolor='gra
#axes[1,1].hist(iris.hist,bin=10)
axes[1,1].hist(iris.PetalLengthCm,bins=100)
#k2.set(xlim=(-1,0.8))
plt.show()
```



Stacked Histogram

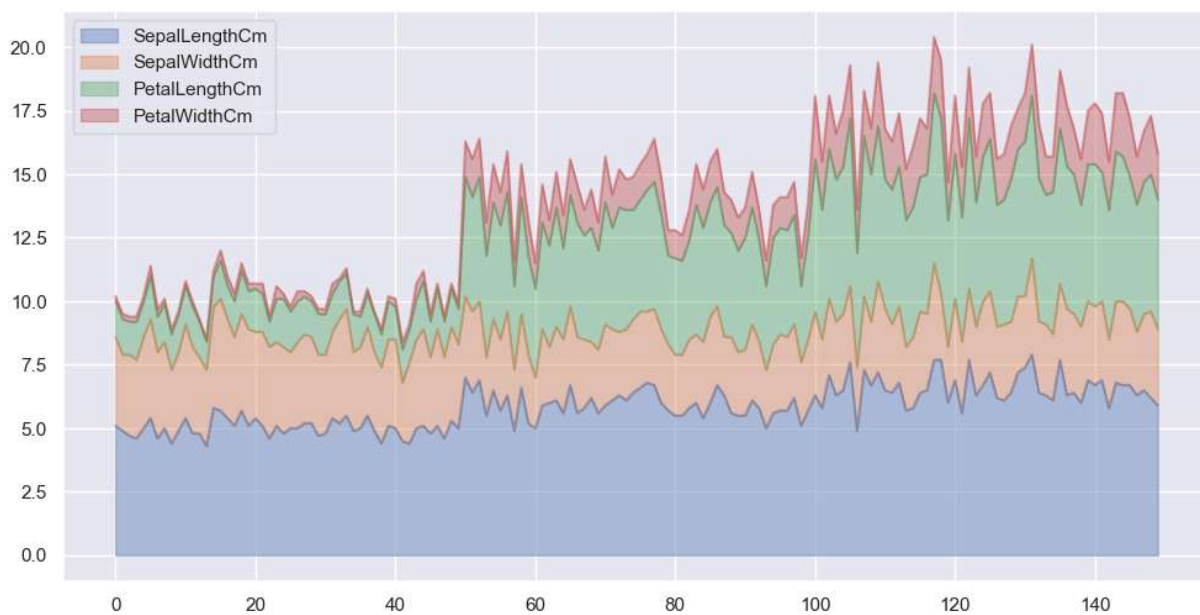
In [52]:
```
iris['Species'] = iris['Species'].astype('category')
```

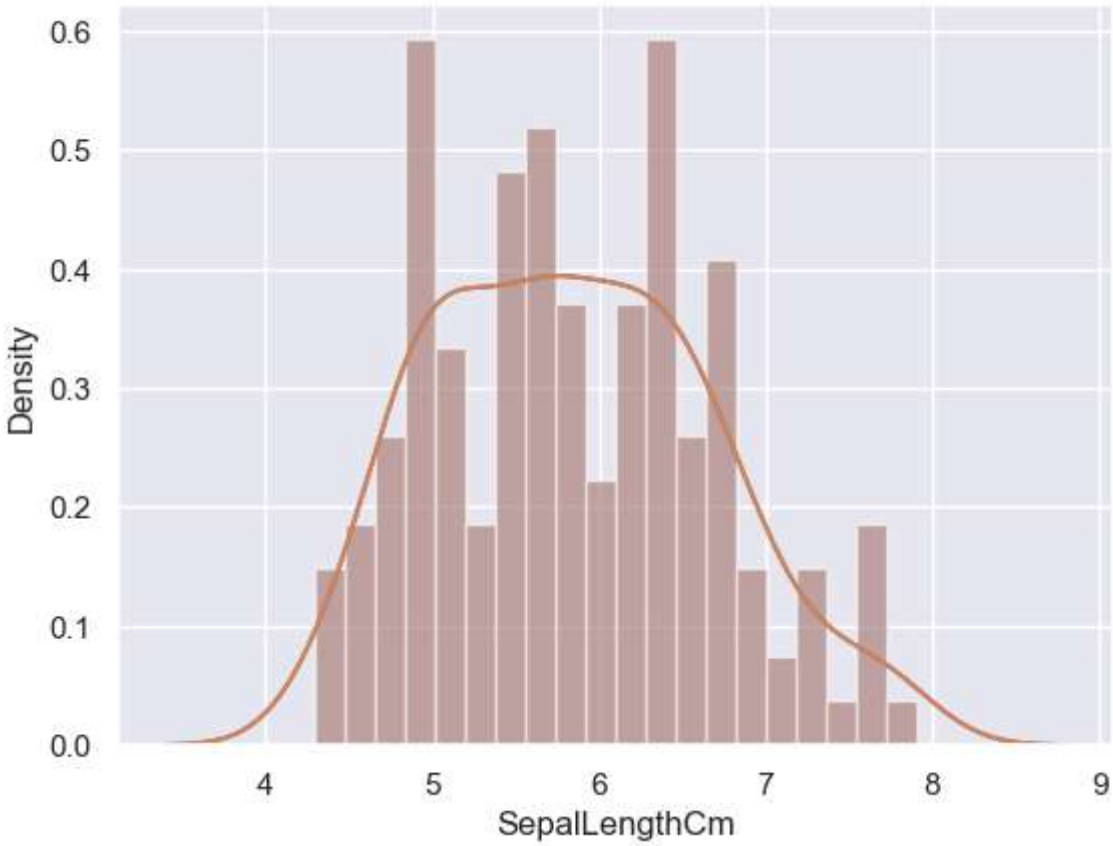
In [53]:
```
list1=list()
mylabels=list()
for gen in iris.Species.cat.categories:
    list1.append(iris[iris.Species==gen].SepalLengthCm)
    mylabels.append(gen)

h=plt.hist(list1,bins=30,stacked=True,rwidth=1,label=mylabels)
plt.legend()
plt.show()
```



Area Plot

In [57]:
```
#iris['SepalLengthCm'] = iris['SepalLengthCm'].astype('category')
#iris.head()
#iris.plot.area(y='SepalLengthCm',alpha=0.4,figsize=(12, 6));
iris.plot.area(y=['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm'],al
plt.show()
```



Distplot

In [59]:
```
sns.distplot(iris['SepalLengthCm'],kde=True,bins=20);
plt.show()
```